

fh6parse Linux kiosk manual

Operator sheet (Polish, daily use only): [LINUX-KIOSK-PL.md](#). This file is the full English install / wiring / update manual.

Version 1.4.0. Raspberry Pi 5 kiosk: portrait **800×600**, MUNBYN **P047**, USB /dev/usb/lp0 print, isometric **line-art** STEP (stick .stp first), **D** vs T warnings, programmed cycle time and a per-tool share chart. Highlight a file to preview ops, cycle time, 3D ready, and the stacked isometric before print. Add mills in **settings** (**Add mill...**: name, rapids m/min, B/C, tool-change time, optional G53 ATC / G54 / travel). Tickets then show a labeled G53 rectangle of where G54 may sit, and max Ø for a centred outside G41/G42. Screen language is **Polish** by default (**F2** / **C** / the **PL-EN** chip for English). A **wireframe 3D cube** next to a file means the STEP views are ready. The screen shows **v1.4.0** (later git commits keep that number; the yellow **UPDATE** button then shows a short hash). The Windows office exe uses the same button against a GitHub Release (tag **vX.Y.Z** builds the exe).

Python **3.10+** is required (Bookworm ships 3.11). Use **Raspberry Pi OS 64-bit Desktop** (Bookworm or later). Pi 5 has no 32-bit OS.

The 40-pin header uses the **same BCM numbers as Pi 3/4**. GPIO on Pi 5 goes through the **RP1** chip: `RP1.GPIO` does **not** work. The kiosk uses `gpiozero` with **lgpio**. Still switch the desktop to **X11** (tkinter + xset blanking).

- 1. [What you need](#)
- 2. [Hardware](#)
- 3. [Software](#)
- 4. [Boot to kiosk](#)
- 5. [Daily use](#)
- 6. [Troubleshooting](#)
- 7. [Files on disk](#)
- 8. [Updating \(1.4.0\)](#)
- 9. [Field test](#)

1. What you need

Item	Notes
Raspberry Pi 5	Official 27 W USB-C PSU (5 V / 5 A). Do not use a Pi 3 2.5 A supply. Do not power the printer from the Pi USB. Active cooler recommended in a closed enclosure.
micro-HDMI cable	Pi 5 has two micro-HDMI ports. Use HDMI0 (the port next to USB-C power) for the kiosk panel.
800×600 LCD, mounted vertically	After rotation the framebuffer is 600×800 . That is what the app uses.
KY-040 rotary encoder	CLK and DT only. The shaft push-switch is unused.
Two momentary buttons	Normally-open, wired to GPIO and GND.
USB stick	FAT/exFAT/NTFS. Programs as .nc / .NC / .tap in the stick root only (not subfolders).

Item	Notes
MUNBYN P047 (ITPP047)	USB, 80 mm ESC/POS, auto-cutter. Own mains PSU.
Company STEP folder (optional)	NAS of .stp / .step if the stick has none. See §3.7 . Stick copy is enough.
Network (optional)	Only for git install and later UPDATE . Printing works offline.
Keyboard / mouse	First-time setup, settings (language, mill / Add mill... , GPIO pins, encoder ticks), SSH, or tap UPDATE . Not required for encoder + GPIO print.

Default GPIO (**BCM** numbers, not header pin numbers):

Function	BCM GPIO	Header pin
Encoder CLK	17	11
Encoder DT	27	13
Full-report button	22	15
Minimal-report button	23	16
3.3 V for encoder VCC	—	1 or 17
GND	—	6, 9, or 14

Change pins in **settings** (**F2** / **C** / **PL-EN**) or in `/etc/fh6parse-kiosk.ini`. Settings write `~/.config/fh6parse/ui.ini` (and the main ini if it is writable).

2. Hardware

2.1 Power and USB

- Pi 5 on the official USB-C 5 V / 5 A supply. USB-C on the Pi is **power only**.
- P047 on its own supply; USB cable to a Pi **USB-A** port is **data only** if the printer has a separate PSU.
- USB stick in any USB-A port (USB 2 or USB 3). Automount under `/media/<user>/...` Or `/run/media/...` is enough; the kiosk polls those paths.

2.2 GPIO rules

Pi GPIO is **3.3 V** (unchanged on Pi 5). Do not feed 5 V into CLK, DT, or the button pins.

KY-040 **VCC** → **3.3 V** (header pin 1), **GND** → **GND**. Many modules work at 3.3 V. If the module insists on 5 V, you still must not put 5 V on the Pi inputs (use a level shifter).

Buttons: one side to the GPIO, the other to GND. The app enables the internal pull-up, so the pin reads high until the button shorts it to ground.

Encoder **SW** (shaft click): leave unconnected.

2.3 Wiring diagram (defaults)

Pi 5 40-pin header (same BCM layout as Pi 3/4), looking at the board with the USB-A / Ethernet ports down:

3.3V	(1)	(2)	5V	← encoder VCC to pin 1 only
GPI02	(3)	(4)	5V	
GPI03	(5)	(6)	GND	← shared GND (encoder GND + both buttons)
GPI04	(7)	(8)	GPI014	
GND	(9)	(10)	GPI015	
GPI017	(11)	(12)	GPI018	← encoder CLK
GPI027	(13)	(14)	GND	← encoder DT
GPI022	(15)	(16)	GPI023	← FULL button MIN button

KY-040 typical labels:

KY-040	Pi
GND	pin 6 (GND)
+	pin 1 (3.3 V)
SW	not used
DT	pin 13 (GPIO 27)
CLK	pin 11 (GPIO 17)

Each print button:

GPIO — button — GND

If turning the knob moves the highlight the wrong way, use **Reverse** in settings (or `encoder_swap = true`, or swap CLK and DT). **Ticks per tooth** is GPIO ticks from one rest valley to the next. The list changes halfway (a 36-tooth knob is 10° per file, ~5° to change the highlight), so a small wiggle at rest does not skip files.

2.4 Screen orientation

The panel is 800×600 landscape electronics, mounted as portrait. The OS must present **600×800**.

On Raspberry Pi OS the default is **Wayland**. Switch to **X11** (tkinter + screensaver `xset` are unreliable on Wayland/labwc):

```
sudo raspi-config
```

- **Advanced Options** → **Wayland** → **X11**
- **System Options** → **Boot / Auto Login** → **Desktop autologin**
- Reboot

Then rotate. Either:

Screen Configuration (desktop): HDMI output → Orientation **right** or **left** until the picture matches the physical panel. Apply and “OK”.

Or in a terminal after login:

```
xrandr --output HDMI-A-1 --rotate right
```

(Use xrandr with no arguments to see the output name: HDMI-A-1, HDMI-1, Pi 5 KMS is usually HDMI-A-1 for HDMI0.)

To make rotation survive reboot, add the same xrandr line to ~/.config/autostart/ or /etc/xdg/lxsession/LXDE-pi/autostart.

Do not use legacy display_rotate= in /boot/firmware/config.txt. Pi 5 is KMS-only.

2.5 Printer (MUNBYN P047)

USB to the Pi. The kiosk sends **raw ESC/POS**: optional STEP bitmaps (gs v 0), then 48-column Font A, then cut. Do **not** print HTML or use a Windows GDI/POS-80 raster driver.

The app writes that blob to /dev/usb/lp0 first (same as open("/dev/usb/lp0", "wb").write(...)). CUPS is not involved on a working Pi. Named CUPS (lp -d munbyn -o raw) is only a fallback if the USB node is missing or busy. There is no lp default-queue attempt.

If CUPS has already claimed the printer, the USB write fails with *Device or resource busy*. Stop or disable that queue (or CUPS) so usb1p owns /dev/usb/lp0.

Before FULL/MIN the kiosk sends **DLE EOT** (real-time status, not printed) on that same node. Cover open, paper end, or cutter error stay on the status line (**Pokrywa otwarta / Brak papieru / Zacięcie drukarki**) and the ticket is not sent. If the firmware does not answer, print goes ahead as before.

3. Software

Do this on the Pi, as user **kiosk**, with network.

3.1 Packages

```
sudo apt update
sudo apt install -y git python3 python3-pip python3-tk \
    python3-gpiozero python3-lgpio python3-rpi-lgpio \
    x11-xserver-utils
```

Do **not** install python3-rpi.gpio on a Pi 5. That library talks to the old SoC GPIO; Pi 5 GPIO is on RP1. python3-lgpio is the driver; python3-rpi-lgpio is only a compatibility shim. The kiosk prefers lgpio automatically.

python3-tk is the kiosk UI. **Do not install CUPS** for the shop P047 — the kiosk prints by writing /dev/usb/lp0. x11-xserver-utils provides xset so the panel can blank.

Confirm Python is 3.10 or newer:

```
python3 --version
```

3.2 Install fh6parse

Recommended — from source (git). This is the shop update path.

```
cd /home/kiosk
git clone https://github.com/Don-Pablo-G/fh6parse.git
cd fh6parse
sudo pip3 install -e . --break-system-packages
```

For isometric views on the ticket, also install the CAD extra (**\$3.7**). On 64-bit Pi 5 this usually has wheels. If cascadio fails, skip it; tickets stay text-only.

```
sudo pip3 install -e '[models]' --break-system-packages
```

--break-system-packages is normal on Bookworm when you are not using a venv. gpiozero stays the **apt** copy so it can see the Pi GPIO.

Allow user kiosk to restart the kiosk after an on-screen **UPDATE** (once, as root):

```
echo 'kiosk ALL=(root) NOPASSWD: /usr/bin/systemctl restart fh6parse-kiosk' | sudo tee
/etc/sudoers.d/fh6parse-kiosk
sudo chmod 440 /etc/sudoers.d/fh6parse-kiosk
```

Check:

```
python3 -m fh6parse --version
```

Expect fh6parse 1.4.0. If the number is older, this clone is behind — `git fetch && git pull --ff-only` then check again (**\$8**).

Later upgrades are **\$8**. Do not run `pip install` on every pull. The kiosk does not update by itself.

Air-gap first copy — prebuilt one-file (this repo's dist/packages). Not the upgrade path; use git for later updates.

On a Pi 5 use the **aarch64** tarball only (there is no 32-bit Raspberry Pi OS for Pi 5).

```
cd /home/kiosk
tar -xzf fh6parse-*-raspberrypi-aarch64.tar.gz
chmod +x fh6parse
sudo cp fh6parse-kiosk.ini.example /etc/fh6parse-kiosk.ini
```

Run:

```
./fh6parse --kiosk --config /etc/fh6parse-kiosk.ini
```

The one-file ARM tarball does not bundle the CAD stack, so **\$3.7** pictures are git-checkout only. There is no **UPDATE** button and --update refuses this install. To get the 1.4.0 shop update path later, switch to the git checkout above.

For systemd, set ExecStart=/home/kiosk/fh6parse --kiosk --config /etc/fh6parse-kiosk.ini (path to the unpacked binary).

3.3 Kiosk config

```
sudo cp /home/kiosk/fh6parse/packaging/fh6parse-kiosk.ini.example /etc/fh6parse-kiosk.ini
sudo nano /etc/fh6parse-kiosk.ini
```

Leave the defaults unless your wiring or printer queue differs. Useful keys:

Key	Default	Meaning
idle_seconds	60	Black screen after this many seconds. 0 disables.
encoder_clk / encoder_dt	17 / 27	BCM pins. Change in settings or here.
button_full / button_min	22 / 23	BCM pins for FULL / MIN. Change in settings or here.
encoder_swap	false	Reverse knob direction
encoder_steps	1	GPIO ticks from one tooth valley to the next. Highlight changes at half a tooth so rest is stable.
printer_device	/dev/usb/lp0	USB printer node (tried first)
printer_queue	(empty)	Optional CUPS name; used only if the USB node fails. Empty = never call lp.
scan_depth	1	USB root only. Raise to search subfolders.
extensions	.nc, .tap	File types (case-insensitive)
extra_roots	(empty)	Extra folders to list, comma-separated (for testing)
model_roots	(empty)	Optional company .stp folders. USB stick is searched first. See §3.7 .
language	p1	Screen language: p1 (default) or en. Change on the kiosk in settings (F2 / C , or click PL / EN). GPIO pins, mill, and encoder ticks are on the same panel. Stored in ~/.config/fh6parse/ui.ini (and in this ini if it is writable).
machine	default	Id of the mill used for cycle time ([machine.<id>] below). Change in settings .
fullscreen	true	Shop display. Escape once exits fullscreen.

Add one [machine.<id>] section per mill, or use **Add mill...** on the Windows/Linux GUI and in kiosk **settings**. Built-in **Default mill** is 20 m/min rapids and 0 s tool change until you pick another. Keys:

Key	Default	Meaning
name	the id	Label on the kiosk, Windows GUI, and tickets
rapid_mm_min	20000	Linear G0 rate (mm/min) for XYZ time
rotary_deg_min	5400	B/C G0 rate (deg/min)
tool_change_s	0	Seconds added at every Txx M6 (carousel swap only)
atc_x atc_y atc_z	(omit)	That mill's tool-change position in G53 mm. Optional atc_b atc_c. Each mill has its own numbers.

Key	Default	Meaning
g54_x g54_y g54_z	(omit)	Typical vise/table part zero in G53 mm for that mill. Optional g54_b g54_c.
tool_length_mm	0	Approximate stick-out (Z only). Haas H is a register, not mm.
x_min x_max y_min y_max	(omit)	Machine travel envelope in G53 mm. Tickets then show where G54 may sit so programmed XY stays inside travel. Optional z_min z_max.

If ATC and G54 XYZ are all set, cycle time uses one G53 pose: work rapids convert through G54+length; at Txx M6 the finishing tool rapids Z then XY/BC to that mill's ATC, then tool_change_s on the new T. A program that already g53's to the ATC is not charged twice. Omit the keys to keep the older estimate (Default mill).

If XY travel is set, the ticket adds a labeled rectangle of allowed G54 origin (G53 mm): four corners, center, max Ø for centred outside G41/G42 (the shorter leftover), and a warning if the stored G54 is outside or the work is larger than travel. Needs Pillow for the PNG; numbers still print without it.

Example:

```
machine = vf-4ss

[machine.vf-4ss]
name = Haas VF-4SS
rapid_mm_min = 25400
rotary_deg_min = 5400
tool_change_s = 2.8
atc_x = -750
atc_y = -20
atc_z = 0
g54_x = -400
g54_y = -250
g54_z = -400
tool_length_mm = 120
x_min = -1270
x_max = 0
y_min = -508
y_max = 0
z_min = -635
z_max = 0
```

3.4 Groups and devices

```
sudo usermod -aG gpio,lp kiosk
```

Log out and back in (or reboot) so the groups apply. Printer node check is **\$3.5**.

3.5 Direct USB printer (recommended)

No CUPS. After the P047 is plugged in:

```
lsusb
ls -l /dev/usb/lp0
```

User kiosk must be able to write the node (lp group, then log out / reboot). Test the same write the kiosk uses:

```
python3 - <<'PY'
from pathlib import Path
Path("/dev/usb/lp0").write_bytes(b"\x1b@TEST\n\n\n\x1dVB\x00")
PY
```

You should get a short slip and a cut. If you see *Permission denied*, the group has not applied yet. If you see *Device or resource busy*, something else (usually CUPS) has the printer — `sudo systemctl stop cups` and try again, then disable CUPS so it does not come back:

```
sudo systemctl disable --now cups
```

If `/dev/usb/lp0` is missing, unplug/replug and look at `dmesg | tail`. A second printer can appear as `/dev/usb/lp1` — set `printer_device` in the ini.

CUPS is not required. Leave `printer_queue` empty to never call lp. Only set a raw queue if the USB node never appears on that machine.

3.6 First run (with keyboard)

On the graphical desktop:

```
python3 -m fh6parse --kiosk --config /etc/fh6parse-kiosk.ini
```

Without GPIO you can still use a **USB keyboard and mouse** at any time (hot-plug is fine). The kiosk keeps keyboard focus and the black screensaver wakes on a key, click, or mouse wheel.

The shop screen is **Polish** unless `language = en` is set. Open **settings** with the keyboard or mouse (not the encoder): **F2** or **C**, or click the **PL / EN** chip next to the version. Pick **Polski** or **English**, the mill (rapids, tool-change time, optional G53 ATC / G54 / tool length / travel from [machine.<id>] in this ini), **Add mill...** to create a new mill (written to `~/.config/fh6parse/ui.ini`), BCM pin numbers for CLK / DT / FULL / MIN, knob reverse, and **ticks per tooth** (GPIO ticks from one rest valley to the next; the highlight changes halfway so a wiggle at rest does not skip files). Language, mill, and GPIO are written to `~/.config/fh6parse/ui.ini` (user kiosk can write this even when `/etc/fh6parse-kiosk.ini` is root-owned) and, if permitted, into the main ini. Pin changes take effect immediately (GPIO is reopened). **Esc** closes the mill form first, then settings; the next **Esc** still leaves fullscreen. Encoder or a GPIO print button closes settings without printing / skipping a file.

Input	While awake	While screensaver
Arrows, mouse wheel, click a file	Move highlight	First event only wakes
F / M	Print full / min	Ignored (no ticket); another key or click wakes
GPIO FULL / MIN	Print	Ignored (no ticket, stays black)
Yellow UPDATE to ... / U	One tap: pull, then restart kiosk	Wake first, then tap
F2 / C / click PL·EN	Open or close settings (language, pins, encoder ticks)	Wake first

Input	While awake	While screensaver
In settings: arrows / wheel / Polski-English	Switch language	—
In settings: + / -	BCM pins and ticks per tooth	—
Esc	Close settings, else leave fullscreen, then close	Wake, then Esc again leaves fullscreen

Plug in a USB stick with .nc files; the list should fill by itself.

Desktop autostart of the kiosk is in the next section. Until then, Escape leaves fullscreen, Escape again closes the window.

3.7 STEP models on the ticket

Optional. The kiosk looks for a matching .stp / .step **on the USB stick first** (same folder as the .nc, then subfolders on that stick). No NAS is required. Company folders in model_roots are a fallback if the stick has no match.

FULL and MIN tickets can show two opposite **true isometric** views (45° then ~35.3° — look along the cube diagonal), as **visible edges only** (silhouette + sharp creases, no shading). Through-holes draw as ellipses (near rim, and the far rim only where you can see through). The longest 3D axis is laid across the 80 mm width (~512 dots); height is cropped to the part, so a long thin shaft is a thin strip, not a metre of paper. A bulky part is capped (~30 mm of paper per view). Print never waits for a model. After an update, delete /tmp/fh6parse-models so old shaded bitmaps are not reused.

1. USB (usual shop path). Put the STEP file next to the program, or in a subfolder on the same stick:

```
D0134078.nc
D0134078_Rev03.stp
```

or D0134078.nc plus cad/D0134078_Rev03.stp. The kiosk indexes the stick when it is inserted. Matching rules are **§3.7 step 4**. CAD extra (**step 3**) is still required to draw the picture.

2. Optional company CAD folders in /etc/fh6parse-kiosk.ini if the stick has no STEP. Several roots are allowed (comma or : / ;) . Subfolders are searched. A file on the stick always wins over the NAS.

```
model_roots = /mnt/fh6parse-cad
```

fh6parse **only reads** that folder (list .stp / .step, open for the isometric). It never creates, overwrites, or deletes files there. Rendered bitmaps go to /tmp/fh6parse-models on the Pi (or %TEMP%\fh6parse-models on Windows). Reports are not written next to company files. CIFS is mounted **read-only**, so the kernel also refuses writes. Network shares (CIFS/NFS) are never treated as a USB stick.

Easy mount of the office Z: drive (plug the Pi into company Ethernet). On a Windows PC, net use Z: shows the UNC (example \\fileserver\Dokumentacja). Then on the Pi:

```
sudo apt install -y cifs-utils
sudo bash /home/kiosk/fh6parse/packaging/connect-windows-share.sh //fileserver/Dokumentacja --
guest
```

Domain account (more common):

```
sudo bash /home/kiosk/fh6parse/packaging/connect-windows-share.sh //fileserver/Dokumentacja --
user kiosk --domain COMPANY
sudo nano /etc/fh6parse-cad.cred
```

Put the password on the password= line, chmod 600, then sudo mount /mnt/fh6parse-cad. The script writes a **read-only** fstab line with x-systemd.automount: when the cable is unplugged the kiosk still starts; when you plug in, the share appears on first use (STEP cubes fill in within a minute). Template: packaging/fh6parse-cad.cred.example.

Manual equivalent:

```
sudo mkdir -p /mnt/fh6parse-cad
# /etc/fstab (one line):
# //fileserver/Dokumentacja /mnt/fh6parse-cad cifs credentials=/etc/fh6parse-
cad.cred,ro,uid=kiosk,gid=kiosk,iocharset=utf8,file_mode=0444,dir_mode=0555,_netdev,nofail,x-
systemd.automount 0 0
```

If the share is down, the ticket is still text only.

On **Windows**, pick **STEP folders...** once. Prefer the UNC (\\fileserver\Dokumentacja) so it still works if the letter is not Z:. That path is model_roots in fh6parse-kiosk.ini. The GUI will not save reports into that folder.

3. Install the CAD extra (git checkout only; heavy: numpy, pillow, trimesh, cascadio):

```
sudo pip3 install -e '[models]' --break-system-packages
```

If that install fails, leave it off. Matching still runs; nothing is rendered and the list icon stays off.

4. Matching starts at the first characters of the NC file name (and the 0 program title). The end of the STEP name may differ (_Rev03, _0P1, extra words). Nearby part numbers do not match (D0134078 will not pick D0134079).

Revision is taken from the G-code header, then the title, then the file name:

In the NC	What is used
(REV 3), (REV.03), (REVISION A)	that revision
(Rewizja: 02), (REW 2), (WERSJA 1)	that revision
001282 (SE0241282-0 WIERCENIA ...)	part SE0241282, rev 0
D0134078_Rev03.nc / D0134078-R2.nc / D0134078A.nc	suffix on the name
No rev in header or name	latest matching .stp / .step

If the program **has** a revision, only a STEP file with the **same** rev is used (not a newer one). If several files share that rev, the closest (shortest) name wins.

NC	Header	Picked model
D0134078.nc	004078 (D0134078) (no REV)	D0134078_Rev03.stp over _Rev02
D0134078.nc	(REV 2)	D0134078_Rev02.stp, not Rev03
SE0241282.nc	001282 (SE0241282-0 ...)	SE0241282-0.stp (or _Rev0)
000814086.nc	title 000814086 OP1/OP2	000814086_Rev02.stp if that is latest

5. On the screen, a small **wireframe 3D cube** appears next to the file when the bitmap is rendered and ready (same icon as the legend under the title — the same visible-edge isometric language as the ticket). The **stacked isometric** itself is shown under the highlight preview (and above the Windows GUI report) so a wrong STEP can be caught before print. The walk and render run in the background for every USB file in the list. Cache: /tmp/fh6parse-models.

On **Windows**, the GUI also searches next to the opened NC file. **STEP folders...** is the company-share fallback (prefer \\server\share, not only Z:). Paths are saved as model_roots in fh6parse-kiosk.ini next to the exe. fh6parse will not write reports into that folder. The Windows one-file build bundles the CAD stack; print still works if a model is missing.

4. Boot to kiosk

Copy the unit and enable it. The service assumes user kiosk, display :0, and Desktop autologin. If model_roots is on a company share, use the read-only automount in **§3.7** (/mnt/fh6parse-cad). The kiosk starts even if the cable is unplugged; cubes appear after the share is up. The sudoers line from **§3.2** must exist if you want the **UPDATE** button to restart the unit.

```
sudo cp /home/kiosk/fh6parse/packaging/fh6parse-kiosk.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable fh6parse-kiosk.service
```

Optional: turn off the *desktop* screensaver so it does not fight the app. In /etc/xdg/lxsession/LXDE-pi/autostart (path may vary) add:

```
@xset s off
@xset -dpms
```

The kiosk still blanks after 60 s via a black overlay and xset dpms force off.

Reboot. After the desktop appears, the CNC list should open fullscreen.

Logs:

```
journalctl -u fh6parse-kiosk -e
```

If the unit starts before X is ready, it will restart every 3 s until :0 exists.

5. Daily use

1. Power on. Screen shows **Włóż pendrive / Insert USB** (or the last stick if it was already plugged in). The kiosk is Polish unless settings were changed.
2. Insert the USB stick. `.nc` / `.tap` files in the stick **root** appear. The count line shows **Można wyjąć / Safe to remove** when the kiosk is not reading the stick — wait for that before unplugging. **Czytanie pendrive — czekaj / Reading USB — wait** means preview parse or a STEP copy from the stick is still running.
3. Turn the encoder to highlight a file. The panel under the list shows each operation's tool count and cycle time, and whether the STEP views are ready. A **3D cube** next to the name also means the STEP views are ready; the stacked isometric appears under the preview when it is. Check OP1 vs OP2 here before printing. If that `.nc` is overwritten on the stick (same name, new bytes), preview reloads from disk — wait for **Reading...** to finish before FULL/MIN.
4. **FULL** — 80 mm ticket: stacked line-art isometrics when ready, then a G54 origin rectangle in G53 mm when the mill has travel (corners, center, max Ø for centred outside G41/G42), operations, cycle time, a share chart of each T, tool list, each tool change (time and % of cycle), warnings, min Z.
5. **MIN** — short ticket: stacked line-art isometrics when ready, then the same G54 rectangle when travel is set, per operation cycle time, share chart of each T, then T, H/D/S to load, description, min Z, and H/D/G95 mismatch flags.
6. If the status line says **Pokrywa otwarta / Printer cover open, Brak papieru / No paper**, or **Zacięcie drukarki / Printer jam**, fix the P047 first. FULL/MIN will not cut a blank slip. Those messages also appear on their own while idle (polled about every 2 s).
7. If there is no cube, print anyway. The slip is text only. The chip next to the cube legend says why: **3D ready, searching...** / **szuka...** (looking for a `.stp`), **rendering...** / **liczy...** (drawing a match), **no STEP** / **brak STEP**, **Z: off** / **Z: wyłącz.** (company share down), or **no CAD** / **brak CAD** (install `[models]`).
8. Status after a good print: **device:/dev/usb/lp0**. If it says `lp:...`, CUPS took the job — **\$3.5**.
9. **WARNING:** lines: `H{n}` does not match `T{tool}` on G43, `D{n}` does not match `T{tool}` on any D, G95 still active... if feed-per-rev was not cancelled with G94 before the next tool (or M30), and empty-pocket (`Txx M6` with no motion) except the **last** tool change (spindle prep). Matching H/D stay quiet. Comments with `!` print as **Programmer notes**.
10. After **60 seconds** with no encoder movement and no new USB, the screen goes black.
11. Wake: encoder, inserting a USB stick, or a **keyboard / mouse**. The first encoder step, key, or click only wakes; it does not skip a file or print. GPIO print buttons while asleep stay ignored.
12. Settings: **F2 / C** or the **PL/EN** chip (mouse) — **\$3.6**. Encoder does not open settings. Mill picker, **Add mill...**, pins, and ticks per tooth are on that panel.
13. Print buttons **do nothing** while the screen is asleep (avoids accidental tickets).
14. Current version is **v...** at the top right. If the Pi is on the network and origin is ahead, a yellow **UPDATE to x.y.z** (or a git hash if the number is still 1.4.0) appears **after this boot's check**. It does **not** update by itself. One tap installs and **restarts** the kiosk (sudoers in **\$3.2**). Print still works until you tap it.

Preview parse runs when a file is highlighted (idle, not on FULL/MIN). STEP matching and rendering run in the background and must not delay the ticket.

6. Troubleshooting

Symptom	What to check
GPIO off: ... on the status line	python3-gpiozero and python3-lgpio installed (not python3-rpi.gpio on Pi 5); user in group gpio; pins not already claimed.
Knob does nothing	CLK/DT on the BCM numbers shown in settings (defaults 17/27); common GND; 3.3 V VCC. Try Reverse .
Knob skips or jitters at rest	Raise Ticks per tooth so the valley is several GPIO ticks wide; highlight only changes halfway to the next tooth. Shorter wires; module decoupling. The kiosk also sets a short encoder bounce_time for Pi 5.
Buttons print on press and release	Use momentary NO to GND, not a latching switch.
List stays on Insert USB / Włóż pendrive	Stick mounted? ls /media / ls /run/media. Format FAT32. Files ending .nc or .tap.
Preview does not match the stick file	Same name overwritten? Wait until Reading... clears. Reload uses mtime and size (FAT 2 s). Do not yank during Reading USB — wait .
Yanked stick, list frozen / cube stuck	Wait for Safe to remove next time. Plug back in. rm -rf /tmp/fh6parse-models if pictures are from a half-copy.
printer failed	ls -l /dev/usb/lp0; user kiosk in group lp; test the Python write in \$3.5. Permission denied → log out after usermod. <i>Busy</i> → CUPS still owns the printer (sudo systemctl disable --now cups).
Brak papieru / Pokrywa otwarta / Zacięcie drukarki	Load paper, close the cover, or clear the cutter. FULL/MIN is blocked on purpose. If the P047 never answers DLE EOT, print still goes through (no false alarm).
Blank slip / cut with no text	Paper was likely already out or the cover was open <i>before</i> this build. Confirm the status line.
Garbage on the slip	CUPS grabbed the job. The kiosk writes /dev/usb/lp0 first; disable CUPS (\$3.5). Status must show device:/dev/usb/lp0, not lp:....
Ticket does not cut	Cutter empty/jammed. Status should show Zacięcie drukarki / Printer jam . App already sends ESC/POS cut (GS V) when status is OK.
Screen never sleeps	idle_seconds = 0, or encoder bouncing. Still on Wayland? Switch to X11 so xset works.
Keyboard/mouse do nothing	Plug into the Pi USB-A; X11 picks them up. Click or press a key — the kiosk claims focus. F2 / C opens settings. Esc closes settings, then leaves fullscreen. GPIO print buttons still do not wake the screensaver.
Language resets to Polish after you picked English	Stored in /home/kiosk/.config/fh6parse/ui.ini. Pick English again in settings. UPDATE does not delete that file. Pins, mill, and ticks live in the same overlay.
Add mill... missing / mill list is only Default	Clone is older than this pull. Settings → Add mill.... Saved in ui.ini ([machine.<id>]). Travel keys x_min...y_max print the G54 rectangle.
Screen stays in English and F2 does nothing	Wake first if the screen is black. Click the EN chip next to v.... Encoder never opens settings.
Black screen immediately	Desktop blanking plus app DPMS. Disable LXDE idle blank; keep kiosk idle_seconds = 60.

Symptom	What to check
Wrong aspect / sideways UI	Rotate until xdpinfo (or Screen Configuration) shows 600×800. App geometry is 600×800 fullscreen. Pi 5 output is often HDMI-A-1.
Service dead, UI never starts	echo \$DISPLAY in a desktop terminal should be :0. raspi-config → X11, desktop autologin. journalctl -u fh6parse-kiosk. Unit User= must be kiosk.
Undervoltage / random reboots	Official 27 W PSU. A phone charger or Pi 3 supply is not enough.
Python 3.9	Wrong image. Flash 64-bit Raspberry Pi OS Desktop for Pi 5.
--update says one-file package	This Pi is running the ARM tarball. Copy a new tarball or reinstall from git (§3.2).
--update / fast-forward failed	Uncommitted edits or a diverged branch. See §8.1 . Do not merge on the shop floor.
--update / UPDATE pulled but UI unchanged	sudo systemctl restart fh6parse-kiosk. Missing sudoers: §3.2 .
UPDATE button never appears	Offline, one-file tarball, already up to date. Check is only at kiosk start. Screen should show v1.4.0 .
UPDATE says failed / kiosk did not restart	sudo -n systemctl restart fh6parse-kiosk from user kiosk should succeed after §3.2 . Then sudo systemctl restart fh6parse-kiosk.
No 3D cube next to files	Read the chip next to the cube legend: searching... (looking for .stp), rendering... (drawing a hit — wait), no STEP (no matching .stp on stick or model_roots), Z: off (company share not mounted), no CAD (pip3 install -e '[models]'). Rev mismatch counts as no STEP. Print still works.
Cube shows, ticket has no picture	Status not device:/dev/usb/lp0 (CUPS intercepted). Printer rejected GS v 0. Test text-only first (§3.5).
Pictures vanished after --update	--update runs pip install -e . without [models] when pyproject.toml changes. Re-run sudo pip3 install -e '[models]' --break-system-packages.
Pictures still shaded / grey mush	Old cache. rm -rf /tmp/fh6parse-models and wait for the cube again. The current renderer is black edges on white only.
No WARNING: for a wrong D	Clone is older than this pull. git log -1 --oneline must mention D vs T. D is checked on G43 and on G41/G42.
No WARNING: after tapping / G95	Next Txx M6 (or M30) must still be in G95. A G94 on the next tool's line cancels it.
No empty-pocket WARNING: on an idle T	The last Txx M6 with no motion is spindle prep (quiet). Earlier idle Txx M6 should warn.

CLI without the kiosk (reports next to the NC file):

```
python3 -m fh6parse /path/program.nc
python3 -m fh6parse --format 80mm-min --stdout /path/program.nc
```

7. Files on disk

Path	Role
/home/kiosk/fh6parse	Source checkout (shop update path)
/etc/fh6parse-kiosk.ini	Pins, printer, idle, model_roots (never overwritten by UPDATE)
/home/kiosk/.config/fh6parse/ui.ini	Screen language, mill (including Add mill...), GPIO pins, encoder ticks. Written by settings. Survives UPDATE .
/etc/systemd/system/fh6parse-kiosk.service	Autostart
/etc/sudoers.d/fh6parse-kiosk	NOPASSWD restart for on-screen UPDATE
packaging/fh6parse-kiosk.ini.example	Template
packaging/fh6parse-kiosk.service	Template
/tmp/fh6parse-models	Cached STEP bitmaps (safe to delete)

8. Updating the kiosk

This section is for **fh6parse 1.4.0** on a **git checkout** (/home/kiosk/fh6parse). The badge stays **1.4.0** until you tag a newer number; shop commits after that badge (preview, mill table / **Add mill...**, travel / G54 box, on-screen isometric, empty-pocket, loops / canned L) still arrive on **UPDATE** as a git hash. Confirm first:

```
python3 -m fh6parse --version
```

You want fh6parse 1.4.0 (also **v1.4.0** on the kiosk). The kiosk **never updates by itself**. Print works with or without a network.

8.1 First pull to 1.4.0 (already installed, older number)

If --version is older than 1.4.0, SSH as kiosk:

```
cd /home/kiosk/fh6parse
git fetch
git pull --ff-only
python3 -m fh6parse --version
```

Add the sudoers line from **§3.2** if it is missing, then:

```
sudo systemctl restart fh6parse-kiosk
```

From this restart onward, later upgrades use **§8.2**.

If git pull --ff-only fails, the clone has local edits or a diverged branch. git status. Do not merge on the shop floor.

8.2 On-screen UPDATE (1.3.2 and later)

On each kiosk start, a background thread runs `git fetch` (~20 s timeout) and compares HEAD to the tracked branch (@{upstream}, else origin/HEAD, else origin/master / origin/main). **Nothing is installed until you tap the button.**

After the check	What you see
Offline, timeout, one-file binary, or already current	No button. v... stays at the top. Print as usual.
Origin has a newer commit	Yellow UPDATE to x.y.z (or a short git hash if the number did not change). Status: <code>v1.4.0 → x.y.z · tap UPDATE to install and restart.</code>

Tap **UPDATE** once (touch or **U**). That is the only action: `git pull --ff-only`, pip only if `pyproject.toml` changed, then **restart** `fh6parse-kiosk`. Print is paused only while that runs. The new version is live after the restart.

kiosk cannot restart a system unit unless you installed the sudoers file in **§3.2**. Without it the pull may succeed and the screen stays on the old process — then:

```
sudo systemctl restart fh6parse-kiosk
```

The check runs **once per start**. After you put a new commit on GitHub, reboot or restart the kiosk (or wait until the next power-on) before the button can appear.

8.3 Keyboard / SSH (--update)

Wake the screen if it is black. **Esc** once leaves fullscreen, **Esc** again closes the window if you need a desktop terminal. If systemd owns the display, open a terminal on `:0` or SSH as kiosk.

```
python3 -m fh6parse --version
python3 -m fh6parse --update
python3 -m fh6parse --version
```

--update and the on-screen button do this, in order:

1. `git pull --ff-only` in the clone (refuses messy merges)
2. `pip3 install -e .` **only if** `pyproject.toml` changed; otherwise skips pip. That command does **not** reinstall the `[models]` extra; see **§3.7** if pictures disappear.
3. `systemctl restart fh6parse-kiosk` if that unit exists; if that fails, `sudo -n systemctl restart fh6parse-kiosk`. Otherwise it prints "restart the kiosk yourself"

It never writes `/etc/fh6parse-kiosk.ini` or `~/.config/fh6parse/ui.ini`. Pins, printer, idle, `model_roots`, and language stay as you set them.

8.4 One-file ARM tarball

No **UPDATE** button. `python3 -m fh6parse --update` (or `./fh6parse --update`) exits with a message to copy a new tarball or switch to a git clone. That package is a first copy, not the 1.4.0 upgrade path. To convert: follow **§3.2** (git + pip), point systemd ExecStart back to `python3 -m fh6parse --kiosk --config /etc/fh6parse-kiosk.ini`, then **§8.2**.

9. Field test

Take this sheet to the Pi. The kiosk must show **v1.4.0** at the top right.

1. Get this code onto the Pi (user kiosk):

```
cd /home/kiosk/fh6parse
git fetch
git checkout master
git pull --ff-only
git log -1 --oneline
sudo systemctl restart fh6parse-kiosk
```

git log -1 on current master should mention **G54** / travel (or **Add mill** / on-screen isometric if that is an older pull). The on-screen badge is still **v1.4.0**. STEP isometrics, D vs T, USB /dev/usb/lp0, cycle time, and kiosk preview are already in older 1.4.0 commits. Or wait for the yellow **UPDATE** after a restart (check runs once per boot).

Clear old STEP bitmaps:

```
rm -rf /tmp/fh6parse-models
```

If /etc/fh6parse-kiosk.ini still has printer_queue = munbyn, you can leave it (USB is tried first) or comment it out. User kiosk must be in group lp. If the printer is busy:

```
sudo systemctl disable --now cups
```

2. Printer (P047)

Check	Pass
ls -l /dev/usb/lp0 exists, writable by kiosk	
Python test in §3.5 prints TEST and cuts	
FULL ticket status: device:/dev/usb/lp0 (not lp:...)	
Ticket text is readable (PC852 Polish comments), then cut	
Open cover → status Pokrywa otwarta ; FULL does not print	
Paper out → status Brak papieru ; FULL does not print	
Cover closed, paper in → FULL prints and cuts	
No CUPS garbage / doubled jobs	

3. D and H vs tool number

Use a program with a wrong offset, or a known sample (000814086.nc T10 with H25 D25).

Check	Pass
Wrong H on G43 → WARNING: H... does not match T...	

Check	Pass
Wrong D on G43 → WARNING: D... does not match T...	
Wrong D on G41/G42 → same D warning, once	
Matching D (= T) → no D warning	
G95 then next T without G94 → WARNING: G95 still active...	
Idle Txx M6 (not the last change) → empty-pocket warning	
Last Txx M6 with no motion stays quiet (spindle prep)	
(...!...) comments listed as programmer notes	
Highlight preview: ops, cycle time, 3D ready, stacked isometric when ready	

4. STEP views (.stp on the USB stick, or model_roots; CAD extra required)

Check	Pass
Stick has Program.nc + matching .stp (root or subfolder)	
3D cube appears when the model is ready (no NAS needed)	
Stacked isometric on the kiosk preview / Windows GUI (not paper only)	
Two stacked views, part not a flat 45° slab (true isometric)	
Black lines on white — no grey shading	
Through-holes as ellipses, not filled blobs	
Long shaft is a thin strip across 80 mm	
Print without a cube is still text-only, no wait	
Settings (F2 / PL chip): language, mill / Add mill... (optional travel Xmin...Ymax), BCM pins, ticks per tooth; survives restart	
Clicky encoder: rest is stable; highlight changes halfway to the next tooth	

Windows office PC: double-click the exe (or `python -m fh6parse --gui` from a git clone). **Polski / English** radios at the top right (default English). Set **STEP folders.... Add mill...** next to the mill combo (name, rapids m/min, B/C, tool-change seconds, optional G53 ATC / G54 / travel). Last NC folder, report folder, and A4 vs 80 mm are remembered. A wireframe cube means the STEP bitmap is ready; the stacked isometric also appears above the report preview. With mill travel set, the ticket includes the G54 origin rectangle in G53 mm. Windows print is still the browser dialog, not `/dev/usb/lp0`. If GitHub (frozen exe) or origin (git) has a newer build, a yellow **UPDATE** button appears after launch — one click, then the window restarts. Frozen **1.4.0** office boxes only show UPDATE after you tag a **newer** version. Publish by tagging **vX.Y.Z** (GitHub Actions builds it) or `packaging\build_windows.bat` then `packaging\publish_windows.ps1`. The tag must match `_version.py`. Do not overwrite `fh6parse-kiosk.ini` next to the exe.